

DHANASHREE SOMANI

PhD Candidate in Statistics

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PROFESSIONAL SUMMARY

PhD candidate in Statistics with expertise in time series econometrics, stochastic processes, and quantitative modeling. Skilled in Python, R, and C++ for forecasting, risk modeling, and financial data analysis. Published in finance research letters, with experience applying machine learning and statistical inference to market volatility and macroeconomic indicators.

TECHNICAL SKILLS

- **Quantitative Finance & Econometrics:** ARIMA, GARCH and its variants like INGARCH and GARCH-X, VaR, Black-Scholes, Monte Carlo simulations, Itô's calculus.
- **Programming:** Python, R, MATLAB, C++, Java, LaTeX.
- **Statistics & Mathematics:** Time series modelling and analysis, Hypothesis testing, Bayesian Analysis, GLMs, Optimization methods, Probability Theory, Numerical Analysis, Linear Algebra.

EDUCATION

UNIVERSITY OF FLORIDA

GAINESVILLE, FLORIDA

PhD in Statistics

2022-Present

PhD Thesis: Exploring comparative forecasting performance of different sequences of nested models, to understand whether a parsimonious model can yield forecasts of statistically equal accuracy, via different econometric metrics, especially in today's time where large parameter-based models such as Neural nets or LLM are widely used.

UNIVERSITY OF FLORIDA

GAINESVILLE, FLORIDA

MS in Statistics

2022-2024

- **MS project:** Implementation and multivariate extension of the well-known Clark-West Test for Predictive Accuracy in nested models, specifically applied to Vector Autoregressive (VAR) models.
- **Relevant Courses:** Measure and Probability Theory, Regression Analysis, Inference, Optimization, Markov chain Monte Carlo Methods, Survival Analysis.

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH

KOLKATA, INDIA

MS in Mathematics and Statistics (BS-MS Dual Degree Course; INSPIRE Fellow)

2017-2022

- **MS Thesis:** Drift Diffusion Model in the TAFC (Two-alternative forced choice) Paradigm and Bayesian Inference
- **Relevant Courses:** Numerical Analysis, Econometrics, Machine Learning and Network analysis, Linear Algebra, Graph theory, Number theory, Topology.
- **Summer Projects**
 - Jump discontinuities in financial time series data and stochastic calculus
 - Random Graph theory and network analysis

PUBLICATIONS

- Shortages and machine-learning forecasting of oil returns volatility: 1900–2024, *Finance Research Letters*, Volume 79, 2025.
The objective of this paper is to forecast the volatility of the West Texas Intermediate oil returns over the monthly period of January 1900 to June 2024 by utilizing the information content of newspapers articles-based indexes shortages for the United States.
- Supply bottlenecks and machine-learning forecasting of international stock market volatility, *Finance Research Letters*, Volume 86, Part G, 2025.
This study explores the information value of the daily Supply Bottlenecks Index – derived from newspaper articles – to forecast daily return volatilities of stock markets of seven major economies – China, France, Germany, Italy, Spain, the UK, and the US.

CERTIFICATIONS AND OTHER COURSES

- Quant Finance Boot Camp: Certificate Program by the Erdős Institute.

PROFESSIONAL EXPERIENCE

UNIVERSITY OF FLORIDA

GAINESVILLE, FLORIDA

Instructor of Record

2023-Present

Designed and taught Engineering Statistics Course for classes of 70+ undergraduate students over 2 semesters. It is a foundational statistics course covering topics like probability, discrete and continuous random variables, confidence interval estimation, hypothesis testing, correlation, regression, and analysis of variance.

Graduate Teaching Assistant

2022-Present

Teaching Assistant for various undergraduate courses like Statistical methods in research, Multivariate statistical analysis, Design of experiments, Statistical modelling, etc.

EXTRA-CURRICULAR

- Received DST-SHE INSPIRE scholarship (2017) - offered to top 1 percentile students in the boards.
- Event co-organizer at college fest – Inquivesta 2018.
- Represented IISER-K in basketball, volleyball and kho-kho at Inter IISER Sports Meet.